Problem sheet 4 for Rigid Analytic Geometry Summer Term 2026

Let K be a field with an absolute value $|\cdot|$ satisfying the conditions from sheet 1. On the previous two sheets we have seen that for any finite extension L/K, this absolute value has a unique extension to an absolute value on L satisfying the same requirements. In the case of an algebraic extension L/K this also holds but L will only be complete if $[L:K] < \infty$. In all cases, the extension will also be denoted $|\cdot|$ and we put $L^\circ = \{l \in L \mid |l| \leq 1\}$. From the uniqueness of the extension it follows that $|\sigma(l)| = |l|$ for every automorphism σ of L/K.

If L/K is finite and $x = (x_i)_{i=1}^m \in L^{\circ m}$, then

$$f(x) = \sum_{\alpha \in \mathbb{N}^m} f_\alpha x^\alpha$$

converges in L for all $f \in \mathbb{T}_m$, the ring of all $f \in K[\![X_1, \ldots, X_m]\!]$ such that $\lim_{|\alpha| \to \infty} |f_\alpha| = 0$. We obtain a ring homomorphism

$$\mathbb{T}_m \xrightarrow{f \to f(x)} L$$

whose image is easily seen to be the sub-K-algebra of L generated by the x_i, which is a field. The kernel

$$\mathfrak{m}_x = \{f \in \mathbb{T}_m \mid f(x) = 0\}$$

is therefore a maximal ideal of $\mathbb{T}_m$. In the lectures, the following will eventually be shown:

Proposition 1. *Every maximal ideal $\mathfrak{m}$ of $\mathbb{T}_m$ has the above form $\mathfrak{m} = \mathfrak{m}_x$, for some finite extension L/K and $x \in L^{\circ m}$.*

In the solutions to the exercise sheets this can be used without proof. Recall from the previous sheets that $\|f\,|\mathbb{T}_m\| = \max\{|f_\alpha| \mid \alpha \in \mathbb{N}^m\}$ defines a Banach algebra norm on $\mathbb{T}_m$ which also defines a valuation on $\mathbb{T}_m$ as $\|fg\,|\mathbb{T}_m\| = \|f\,|\mathbb{T}_m\| \cdot \|g\,|\mathbb{T}_m\|$.

Problem 1 (1 point)**.** *For $x \in L^{\circ m}$ as above and $f \in \mathbb{T}_m$, we have*

$$|f(x)| \leq \|f\,|\mathbb{T}_m\|\,. \tag{1}$$

Problem 2 (5 points)**.** *For $f \in \mathbb{T}_m$, show that the following conditions are equivalent:*

- *$f(x) \neq 0$ for all finite extensions L/K and all $x \in L^{\circ m}$.*
- *$f \in \mathbb{T}_m^\times$*
- *$f \neq 0$ and for all finite extensions L/K and all $x \in L^{\circ m}$, (1) becomes an equality.*

Problem 3 (3 points)**.** *Assume that K is algebraically closed, $x \in K^\circ$, $f \in \mathbb{T}_1$ and $R \in (0,1]_{\mathbb{R}}$ such that f has no zeroes on*

$$K_{\leq R}(x) = \big\{ y \in K \;\big|\; |x-y| \leq R \big\}.$$

Show that $y \to |f(y)|$ is constant on $K_{\leq R}(x)$!

We would also like to show that the completion with respect to $|\cdot|$ of the algebraic closure of K is algebraically closed. An example of this will be $\mathbb{C}_p$, the completion of the algebraic closure of $\mathbb{Q}_p$. Verifying algebraic closedness splits into verifying perfectness and separable closedness where the first task is of course easy.

Problem 4 (2 points)**.** *Let L be an algebraic extension of K which is perfect, and let $\mathbb{L}$ be its completion with respect to $|\cdot|$. Show that $\mathbb{L}$ is perfect!*

For separable closedness, the following observation of Krasner will be useful.

Problem 5 (5 points)**.** *Let L/K be a Galois extension, $\alpha \in L$ and let $\beta \in L$ be such that*

$$|\beta - \alpha| < |\beta - \tilde{\alpha}|$$

when $\tilde{\alpha} \neq \alpha$ is any zero in L of the minimal polynomial $\mathrm{Min}_{\alpha/K}$. Show that $\alpha \in K(\beta)$!

It will also be necessary to have an estimate in terms of the absolute values of polynomial coefficients for the absolute values of the zeros of the polynomial. For complex coefficients such estimates should be well known from the basic analysis lectures as most proofs of the Fundamental Theorem of Algebra require such estimates. In the case of ultrametric absolute values, sharper results are possible and the absolute values of the zeroes can actually be read off the slopes of the Newton polygon of the polynomial. For our purposes, the following is sufficient:

Problem 6 (4 points)**.** *Let d be a positive integer, $(\alpha_i)_{i=1}^d \in K^d$ and $\prod_{i=1}^d (T - \alpha_i) = \sum_{k=0}^d p_k T^k$. Then*

$$\max(|\alpha_1|, \ldots, |\alpha_d|) = \max\Big\{ \sqrt[d-i]{|p_i|} \;\Big|\; 0 \leq i < d \Big\} \tag{2}$$

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday May 18.